LAB 2: Use Case Diagram

Smart Waste Collection Management System

KIET Deemed To Be University | Computer Science Engineering | CSE-E

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1. Objective

To draw the use case diagram of the Smart Waste Collection Management System, to specify the role of each actor, and to state the function, precondition and postcondition of each use case. All items are taken from the Software Requirements Specification (SRS v1.0): user classes (2.3), product functions (2.2), system features (section 4) and business rules (5.5). The diagram is on page 2 and the use case specifications are on pages 3 and 4.

2. Actors and their roles

Actor	Type	Role in the system	Use cases
Citizen	Primary (human)	Member of the public with no technical expertise. Reports overflowing bins or asks for an unscheduled pickup, and follows the status of the complaint.	UC-12, UC-13
Municipal Administrator	Primary (human)	Manages the system from the web dashboard: views bin status and analytics, configures alert thresholds, generates and edits collection routes. Must authenticate.	UC-03 to UC-08
Collection Staff / Driver	Primary (human)	Uses the mobile app with minimal technical effort: views the day's optimized route and marks bins as collected.	UC-09, UC-10
System Administrator	Primary (human)	Technically skilled person who maintains the servers, sensors and connectivity.	UC-14
Sensor Node (IoT device)	Secondary (system)	Ultrasonic sensor with a microcontroller on each bin. Measures the fill-level and sends it to the backend.	UC-01
Push Notification Service (FCM)	Secondary (system)	External service that delivers alerts to the administrator dashboard and the driver's app.	UC-02
Mapping API (OSM / Google Maps)	Secondary (system)	External service that supplies map and distance data used for route optimization.	UC-07

3. Relationships between use cases

Relationship	Meaning
UC-05, UC-06, UC-03, UC-07 <> UC-04 Log in	Every administrator function needs authentication first (REQ-DB-3, SRS 5.3).
UC-10 <> UC-11 Resolve linked complaints	Marking a bin collected always resolves the complaints of that bin (SRS 5.5).
UC-02 <> UC-01 Monitor bin fill-level	The alert is sent only when the fill-level reaches the threshold.
UC-08 <> UC-07 Generate optimized route	Editing the route happens only if the administrator wants to change it (REQ-RO-3).

The business rule that a bin joins a route only when its fill-level is at or above the threshold (SRS 5.5) is part of UC-07. Drivers and citizens use the mobile app; the SRS requires authentication only for administrators, so Log in is shown for the administrator functions only.

Use Case Diagram - Smart Waste Collection Management System

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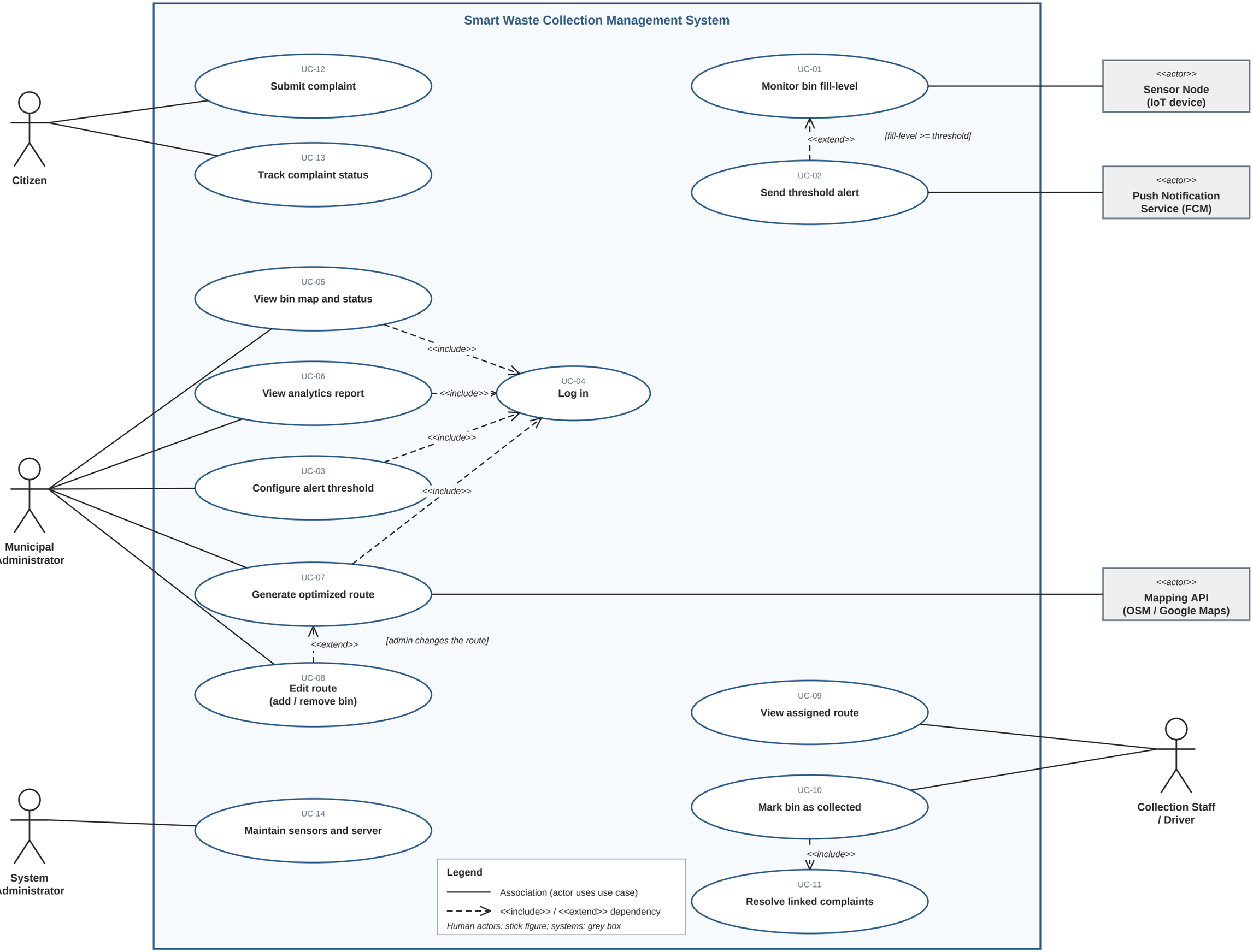

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4. Use case specifications

Use case	Actor	Function	Precondition	Postcondition	SRS ref.
UC-01 **Monitor bin fill-level**	Sensor Node	Measures the distance to the waste surface, converts it to a fill percentage and sends it with the bin ID and timestamp at a configurable interval (default 15 minutes). Failed reads are retried and logged.	Sensor node is installed on the bin, powered and connected over Wi-Fi / GPRS. The bin is registered in the system.	Reading is stored. The bin's fill-level and status are updated on the dashboard. Failed readings are logged.	REQ-BM-1 to 3
UC-02 **Send threshold alert**	Push Notification Service (receivers: Administrator, Driver)	Generates an alert within 5 minutes when a bin reaches its threshold, logs it and sends a push notification.	A new reading is stored. Fill-level is at or above the threshold. No open alert exists for the bin.	Alert is logged with bin ID and timestamp. Notification reaches the administrator dashboard and the driver's app.	REQ-AL-2, REQ-AL-3
UC-03 **Configure alert threshold**	Municipal Administrator	Sets the alert threshold percentage for one bin or for all bins (example default 80%).	Administrator is logged in (UC-04).	New threshold is saved and used for the next readings and for route selection.	REQ-AL-1, SRS 5.5
UC-04 **Log in**	Municipal Administrator	Authenticates with username and password (optional OTP) over HTTPS before the dashboard is opened.	Administrator account exists and the administrator knows the credentials.	On success a session starts and the dashboard opens. On failure access is denied and an error message is shown.	REQ-DB-3, SRS 5.3
UC-05 **View bin map and status**	Municipal Administrator	Shows all bins on a map coloured green, yellow or red by fill-level, with a list of bins that need collection and a drill-down to a bin's history.	Administrator is logged in. At least one bin is registered.	Current status of all bins is displayed. No data is changed.	REQ-DB-1
UC-06 **View analytics report**	Municipal Administrator	Shows the number of collections and the average fill-level per bin for a selected date range.	Administrator is logged in. A date range is selected. Historical records exist.	Report is displayed. Stored records are unchanged.	REQ-DB-2
UC-07 **Generate optimized route**	Municipal Administrator (or daily scheduled trigger); Mapping API	Collects all bins at or above the threshold, computes an optimized route (within 10 seconds for up to 200 bins), shows it on the dashboard and pushes it to the driver's app.	Administrator is logged in (or the daily trigger fires). At least one bin is at or above the threshold. Map data is available.	Route is saved and displayed. Included bins are marked as scheduled. Route is sent to the driver.	REQ-RO-1, REQ-RO-2, SRS 5.5
UC-08 **Edit route (add / remove bin)**	Municipal Administrator	Manually adds a bin to or removes a bin from a generated route; the route is recomputed.	A generated route exists. Administrator is logged in.	Route contains the changed list of bins and the updated route is available to the driver.	REQ-RO-3
UC-09 **View assigned route**	Driver	Shows the day's optimized route as a list or map of the bins to collect.	Driver has the mobile app. A route has been sent to the driver.	Route is displayed. Nothing is changed.	SRS 3.1
UC-10 **Mark bin as collected**	Driver	Records that a bin has been emptied. The bin's fill-level is reset and its alert is closed.	Driver is viewing the assigned route. The bin on the route has been emptied.	Bin is recorded as collected and the collection is available for analytics. Linked complaints are resolved (UC-11).	SRS 3.1, SRS 5.5

Use case	Actor	Function	Precondition	Postcondition	SRS ref.
UC-11 **Resolve linked complaints**	System (included by UC-10)	Automatically changes every Pending complaint of the collected bin to Resolved.	Bin has been recorded as collected. Pending complaints exist for that bin.	Complaints are marked Resolved, and citizens see the new status.	SRS 5.5, REQ-CF-3
UC-12 **Submit complaint**	Citizen	Reports an overflowing bin or requests an unscheduled pickup with the location (auto-detected or selected) and an optional photo. A unique tracking ID is assigned.	Citizen has the mobile app or web form and internet access. A bin location is available or selected.	Complaint is stored with the status Pending and a tracking ID, and is forwarded to the administrator dashboard.	REQ-CF-1, REQ-CF-2
UC-13 **Track complaint status**	Citizen	Shows the current status (Pending or Resolved) of a complaint using its tracking ID.	A complaint has been submitted. Citizen has the tracking ID.	Current status is displayed. Nothing is changed.	REQ-CF-3
UC-14 **Maintain sensors and server**	System Administrator	Keeps the servers, sensors and connectivity working, for example by fixing connections, servicing sensor nodes and keeping the backend running.	System Administrator has technical access to the servers and the sensor hardware.	Backend and sensors are operational and detected faults are corrected.	SRS 2.3, 5.2, 5.4